

1. Administrative & Study Identification

Full Study Title: Safety and Efficacy Study of Pembrolizumab (MK-3475) Combined With Lenvatinib (MK-7902/E7080) as First-line Intervention in Adults With Advance Melanoma (MK-7902-003/E7080-G000-312/LEAP-003) **Short Title/Acronym:** LEAP-003 **Protocol Number:** MK-7902-003 / E7080-G000-312 **NCT Number:** NCT03820986 **Sponsor Name:** Merck **Investigational Product:** Pembrolizumab (MK-3475) + Lenvatinib (MK-7902 / E7080 / LENVIMA, ATC Code: L01EX08) **Phase of Development:** Phase 3 **Medical Monitor:** Clinical Operations Lead / Medical Monitor (Merck)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess whether the combination of pembrolizumab and lenvatinib is superior to pembrolizumab and placebo as assessed by Progression-free Survival (PFS) and Overall Survival (OS). **Secondary Objectives:** To assess the Objective Response Rate (ORR), Duration of Response (DOR), and the number of participants with adverse events (AEs).

2.2 Background & Rationale To provide an alternative, more effective first-line treatment option for previously untreated and unresectable advanced melanoma. Tumor progression and immune evasion result from multiple oncogenic and immunosuppressive signals within the tumor microenvironment. Combining lenvatinib (a multikinase inhibitor targeting VEGF and other pathways) with pembrolizumab (a PD-1 inhibitor) has been shown in preclinical models to enhance immune activation and disrupt tumor blood supply, potentially overcoming resistance to immunotherapy and improving survival outcomes for patients with this difficult-to-treat cancer.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (Pembrolizumab + Placebo) **Blinding:** Double-blind (Note: As of 03-April-2023 active participants, investigator, and sponsor personnel or delegate(s) involved in the treatment administration or clinical evaluation of the participants will be unblinded). **Randomization:** 1:1 Randomized

3.2 Planned Enrollment & Duration Target **SN\$:** 674 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 2019 **Trial Status:** COMPLETED (Study has been completed)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults (≥ 18 years) with histologically or cytologically confirmed unresectable stage III or stage IV melanoma. 2. Must be untreated for advanced or metastatic disease (Note: Patients with BRAF V600 mutation-positive melanoma are allowed to have received standard-of-care targeted therapy as first-line therapy). 3. Has at least 1 measurable lesion by RECIST v1.1 criteria. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Has received prior systemic treatment for unresectable or metastatic melanoma other than permitted targeted therapy. 2. Prior adjuvant or neoadjuvant therapy with targeted therapy or immunotherapy (anti-CTLA-4, anti-PD-1) if relapse occurred during active treatment or within 6 months of treatment discontinuation. 3. Prior therapy with a monoclonal antibody, chemotherapy, or investigational agent within 4 weeks (or 5 half-lives) before study treatment administration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Pembrolizumab 200 mg (or 300 mg every 3 weeks per cohort specifics) via intravenous (IV) infusion on Day 1 of each 3-week cycle PLUS lenvatinib 20 mg via oral capsule once daily. **Route of Administration:** Intravenous (Pembrolizumab) and Oral (Lenvatinib). **EMA Product Information:** Lenvatinib EMA URL: https://www.ema.europa.eu/en/documents/product-information/lenvima-epar-product-information_en.pdf **Duration of Treatment:** Pembrolizumab administered for up to 35 cycles (approximately 2 years). Lenvatinib administered for up to at least 2 years; after 2 years of combination therapy, lenvatinib may be administered as a single agent until protocol-specified discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care to mitigate adverse events. **Prohibited:** Other systemic anti-cancer therapies, live vaccines, and concurrent investigational agents during the study treatment phase.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Medical History, Tumor Imaging (Baseline), Biomarker Testing (BRAF) | | **Baseline** | Day 0 | Randomization (1:1), First IV Infusion, First Oral Dose dispensed | | **Interim** | Every 3 Weeks (Q3W) | IV Infusion, Safety Labs, AE/Concomitant Meds Review, Lenvatinib dispensation. Tumor imaging conducted periodically (e.g., every 9 weeks). | | **End of Study** | Up to 2 Years | End of combination treatment, 30-day safety follow-up, 90-day SAE

follow-up, and long-term survival follow-up until death or study withdrawal. |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) and Overall Survival (OS). **Measurement Method:** Evaluated per Modified Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST 1.1) by Blinded Independent Central Review (BICR). For this study, RECIST 1.1 has been modified to follow a maximum of 10 target lesions and a maximum of 5 target lesions per organ.

7.2 Secondary Endpoints - Objective Response Rate (ORR) as Assessed by BICR per RECIST 1.1 - Duration of Response (DOR) as Assessed by BICR per RECIST 1.1 - Number of Participants With Adverse Events (AEs)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard monitoring collected throughout the study and for 30 days post-treatment termination. **Serious Adverse Events (SAEs):** Evaluated and collected for up to 90 days post-treatment. **Data Monitoring Committee (DMC):** External, independent Data Monitoring Committee (eDMC) established to regularly review safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy endpoints (OS, PFS). Safety Set for all AE/SAE evaluations. **Sample Size Justification:** $N=674$ appropriately powered to detect significant differences in dual primary endpoints between the experimental and placebo arms. **Handling of Missing Data:** Standard survival analysis right-censoring methods applied for OS and PFS endpoints for participants without documented events at cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Supervised by an external data monitoring committee; 3 interim analyses and a final analysis were pre-planned for efficacy and safety bounding. **Audit Trail:** eCRF locking, tracking of protocol deviations, and Blinded Independent Central Review (BICR) for all radiographic scans to ensure bias-free assessment.

11. Study Identifiers & Governance

Primary ID: {{MK-7902-003 / E7080-G000-312}} **Registry Number:** {{NCT03820986}} **Posting ID:** {{725161721349314373}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{LEAP-003}} **Official Regulatory Title:** {{A Phase 3 Randomized, Placebo-controlled Trial to Evaluate the Safety and Efficacy of Pembrolizumab (MK-3475) and Lenvatinib (E7080/MK-7902) Versus Pembrolizumab Alone as First-line Intervention in Participants With Advanced Melanoma (LEAP-003)}}

12. Clinical Framework (Melanoma Specific)

Primary Condition: {{Malignant Melanoma (SNOMED: 372244006 | MeSH: D008545 | HPO: HP:0002861 | Semantic Type: Neoplastic Process)}} **Diagnosis Threshold:** {{Unresectable Stage III or Stage IV}} **Medical Methodology:** {{PD-1 inhibitor + Tyrosine Kinase Inhibitor combination therapy}} **Optimization Target:** {{Progression-Free Survival (PFS) and Overall Survival (OS)}}

13. Study Procedures & Methodology

Management Pathway: {{First-line oncology clinical pathway}} **Education Protocol (HCP):** {{Site initiation and infusion/AE management training}} **Education Protocol (Patient):** {{Oral medication compliance instructions and AE self-reporting}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for radiological assessments}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 24 months of treatment + long-term survival follow-up}} **Onsite Visit Cadence:** {{Every 3 weeks (Q3W) for IV infusion}} **Remote Monitoring Frequency:** {{Safety follow-up calls as needed}} **Checklist Review Interval:** {{Tumor imaging review according to RECIST 1.1 schedule (e.g., every 9 weeks)}}

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Name]		_____		[DD-MMM-YYYY]										
	Clinical Operations Lead		[Name]		_____		[DD-MMM-YYYY]										
	Statistician		[Name]		_____		[DD-MMM-YYYY]										